
Applied Data Mining

— Week 12—
Spring 2026

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Contents

- Hierarchical Clustering

What is Agglomerative Clustering?

- **Agglomerative** means **bottom-up** clustering.
- Initially, every data point is its **own cluster**.
- Then, the **closest clusters are merged repeatedly** until only one cluster remains.

Initially:

$\{a\}, \{b\}, \{c\}, \{d\}, \{e\}$

Single Linkage Agglomerative Clustering

Single Linkage

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

Agglomerative

Clustering

Solved Example

Single Linkage Agglomerative Clustering

Single Linkage Agglomerative Clustering Solved Example

- Given the dataset {a, b, c, d, e} and the following distance matrix,
- Construct a dendrogram by Single - linkage hierarchical clustering using the **agglomerative method**.

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

Single Linkage Agglomerative Clustering

Single Linkage Agglomerative Clustering Solved Example

- The Single-linkage clustering uses the “minimum formula”, that is, the following formula to compute the distance between two clusters A and B:

- $d(A, B)$ = $\min$ $\{d(x, y) : x \in A, y \in B\}$

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

Single Linkage Agglomerative Clustering

Single Linkage Agglomerative Clustering Solved Example

- Let us compute the distance of $\{c, e\}$ from other clusters.
- $d(\{c, e\}, \{a\}) = \min\{d(c, a), d(e, a)\} = \min\{3, 11\} = 3$
- $d(\{c, e\}, \{b\}) = \min\{d(c, b), d(e, b)\} = \min\{7, 10\} = 7$
- $d(\{c, e\}, \{d\}) = \min\{d(c, d), d(e, d)\} = \min\{9, 8\} = 8$
- The following table gives the distances between the various clusters in C2.

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

	a	b	c,e	d
a	0	9	3	6
b	9	0	7	5
c,e	3	7	0	8
d	6	5	8	0

Single Linkage Agglomerative Clustering

Single Linkage Agglomerative Clustering Solved Example

- From the table, the minimum distance is the distance between the clusters **{a}** and **{c, e}**.
- Also, $d(\{a\}, \{c, e\}) = 3$.
- We merge **{a}** and **{c, e}** to form the cluster **{a, c, e}**.
- The new set of clusters **C3: {a, c, e}, {b}, {d}**.

	a	b	c,e	d
a	0	9	3	6
b	9	0	7	5
c,e	3	7	0	8
d	6	5	8	0

	a,c,e	b	d
a,c,e	0	?	?
b	?	0	5
d	?	5	0

Single Linkage Agglomerative Clustering

Single Linkage Agglomerative Clustering Solved Example

- Let us compute the distance of $\{a, c, e\}$ from other clusters.
- $d(\{a, c, e\}, \{b\}) = \min\{d(a, b), d(c, b), d(e, b)\}$
- $d(\{a, c, e\}, \{b\}) = \{9, 7, 10\} = 7$
- $d(\{a, c, e\}, \{d\}) = \min\{d(a, d), d(c, d), d(e, d)\}$
- $d(\{a, c, e\}, \{d\}) = \{6, 9, 8\} = 6$

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

	a,c,e	b	d
a,c,e	0	7	6
b	7	0	5
d	6	5	0

Single Linkage Agglomerative Clustering

Single Linkage Agglomerative Clustering Solved Example

- $d(\{a, c, e\}, \{b, d\}) =$
 $\min\{d(a, b), d(a, d), d(c, b), d(c, d), d(e, b), d(e, d)\}$
- $d(\{a, c, e\}, \{b, d\}) = \min\{9, 6, 7, 9, 10, 8\}$
- $d(\{a, c, e\}, \{b, d\}) = 6$

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

	a,c,e	b,d
a,c,e	0	6
b,d	6	0

Single Linkage Agglomerative Clustering

Single Linkage Agglomerative Clustering Solved Example

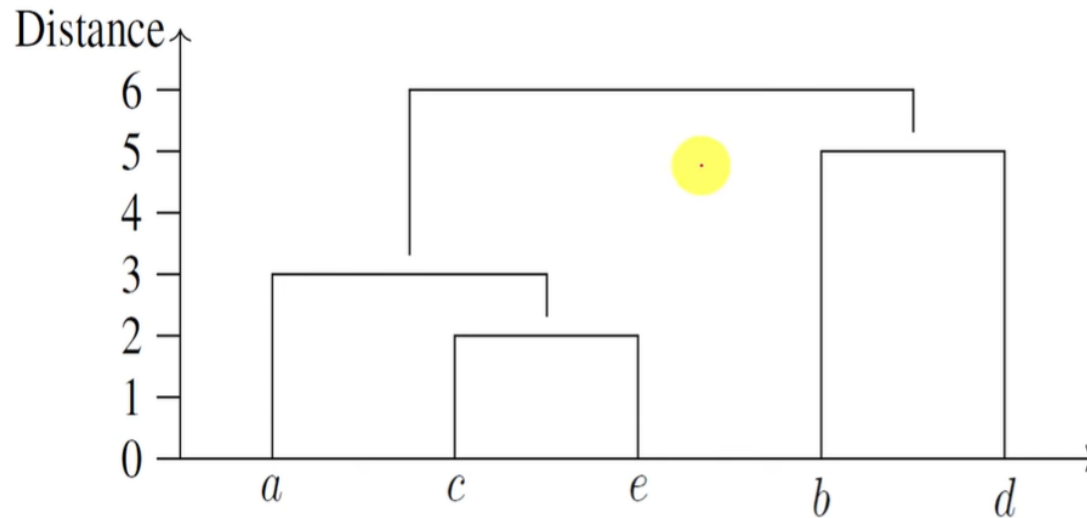
- $d(\{a, c, e\}, \{b, d\}) =$
 $\min\{d(a, b), d(a, d), d(c, b), d(c, d), d(e, b), d(e, d)\}$
- $d(\{a, c, e\}, \{b, d\}) = \min\{9, 6, 7, 9, 10, 8\}$
- $d(\{a, c, e\}, \{b, d\}) = 6$
- Only two clusters are left. We merge them form a single cluster containing all data points.

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

	a,c,e	b,d
a,c,e	0	6
b,d	6	0

Single Linkage Agglomerative Clustering

Single Linkage Agglomerative Clustering Solved Example



	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

Average Linkage Agglomerative Clustering

Average Linkage

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

Agglomerative
Clustering
Solved Example

Average Linkage Agglomerative Clustering

Average Linkage Agglomerative Clustering Solved Example

- Given the dataset {a, b, c, d, e} and the following distance matrix,
- Construct a dendrogram by Average - linkage hierarchical clustering using the agglomerative method.

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

Average Linkage Agglomerative Clustering

Average Linkage Agglomerative Clustering Solved Example

- The Average-linkage clustering uses the “average formula”, that is, the following formula to compute the distance between two clusters A and B:

- $\underline{d(A, B)} = \underline{avg}\{\underline{d(x, y)}: x \in A, y \in B\}$

- $\overset{\checkmark}{d}(\overset{\checkmark}{A}, \overset{\checkmark}{B}) = \frac{\sum d(x, y): x \in A, y \in B}{|A||B|}$

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

number of elements in cluster A & B

Average Linkage Agglomerative Clustering

Average Linkage Agglomerative Clustering Solved Example

- Dataset : $\{a, b, c, d, e\}$.
- Initial clustering (Singleton sets)
- C_1 : $\{a\}, \{b\}, \{c\}, \{d\}, \{e\}$.
- From the table, the minimum distance is the distance between the clusters $\{c\}$ and $\{e\}$.

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

Average Linkage Agglomerative Clustering

Average Linkage Agglomerative Clustering Solved Example

- Dataset : $\{a, b, c, d, e\}$.
- Initial clustering (Singleton sets)
- C1**: $\{a\}, \{b\}, \{c\}, \{d\}, \{e\}$.
- From the table, the minimum distance is the distance between the clusters $\{c\}$ and $\{e\}$.
- Also, $d(\{c\}, \{e\}) = 2$.
- We merge $\{c\}$ and $\{e\}$ to form the cluster $\{c, e\}$.
- The new set of clusters **C2**: $\{a\}, \{b\}, \{d\}, \{c, e\}$.

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

	a	b	c,e	d
a	0	9	?	6
b	9	0	?	5
c,e	?	?	0	?
d	6	5	?	0

Average Linkage Agglomerative Clustering

Average Linkage Agglomerative Clustering Solved Example

- Let us compute the distance of {c, e} from other clusters.

- $d(\{c, e\}, \{a\}) = \text{avg}\{d(c, a), d(e, a)\} = \frac{3+11}{2*1} = 7$
- $d(\{c, e\}, \{b\}) = \text{avg}\{d(c, b), d(e, b)\} = \frac{7+10}{2*1} = 8.5$
- $d(\{c, e\}, \{d\}) = \text{avg}\{d(c, d), d(e, d)\} = \frac{9+8}{2*1} = 8.5$
- The following table gives the distances between the various clusters in C2.

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

	a	b	c,e	d
a	0	9	7	6
b	9	0	8.5	5
c,e	7	8.5	0	8.5
d	6	5	8.5	0

Average Linkage Agglomerative Clustering

Average Linkage Agglomerative Clustering Solved Example

- From the table, the minimum distance is the distance between the clusters **{d}** and **{d}**.

	a	b	c,e	d
a	0	9	7	6
b	9	0	8.5	5
c,e	7	8.5	0	8.5
d	6	5	8.5	0

Average Linkage Agglomerative Clustering

Average Linkage Agglomerative Clustering Solved Example

- From the table, the minimum distance is the distance between the clusters **{d}** and **{d}**.
- Also, $d(\{b\}, \{d\}) = 5$.
- We merge **{b}** and **{d}** to form the cluster **{b, d}**.
- The new set of clusters **C3: {a}, {c, e}, {b, d}**.

	a	b	c,e	d
a	0	9	7	6
b	9	0	8.5	5
c,e	7	8.5	0	8.5
d	6	5	8.5	0

	a	b,d	c,e
a	0	?	7
b,d	?	0	?
c,e	7	?	0

Average Linkage Agglomerative Clustering

Average Linkage Agglomerative Clustering Solved Example

- Let us compute the distance of $\{b, d\}$ from other clusters.

- $\underline{d(\{b, d\}, \{a\})} = \text{avg}\{\underline{d(b, a)}, d(d, a)\}$

- $d(\{b, d\}, \{a\}) = \frac{9+6}{2*1} = 7.5$ ✓

- $d(\{b, d\}, \{c, e\}) =$
 $\text{avg}\{d(b, c), d(b, e), d(d, c), d(d, e)\}$

- $d(\{b, d\}, \{c, e\}) = \frac{7+10+9+8}{2*2} = 8.5$

	a	b	c	d	e
a	0	9	3	6	11
b	9 ✓	0	7	5	10
c	3	7	0	9	2
d	6 ✓	5	9	0	8
e	11	10	2	8	0

	a	b,d	c,e
a	0	?	7
b,d	7 ✓	0	?
c,e	7	?	0

Average Linkage Agglomerative Clustering

Average Linkage Agglomerative Clustering Solved Example

- Let us compute the distance of $\{b, d\}$ from other clusters.

- $d(\{b, d\}, \{a\}) = \text{avg}\{d(b, a), d(d, a)\}$

- $d(\{b, d\}, \{a\}) = \frac{9+6}{2 \cdot 1} = 7.5$

- $d(\{b, d\}, \{c, e\}) =$
 $\text{avg}\{d(b, c), d(b, e), d(d, c), d(d, e)\}$

- $d(\{b, d\}, \{c, e\}) = \frac{7+10+9+8}{2 \cdot 2} = 8.5$

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

	a	b,d	c,e
a	0	7.5	7
b,d	7.5	0	8.5
c,e	7	8.5	0

Average Linkage Agglomerative Clustering

Average Linkage Agglomerative Clustering Solved Example

- From the table, the minimum distance is the distance between the clusters **{a}** and **{c, e}**.
- Also, $d(\{a\}, \{c, e\}) = 7$
- We merge **{a}** and **{b, d}** to form the cluster **{a, b, d}**.
- The new set of clusters **C4: {a, c, e}, {b, d}**

	a	b, d	c, e
a	0	7.5	7
b, d	7.5	0	8.5
c, e	7	8.5	0

	a, c, e	b, d
a, c, e	0	?
b, d	?	0

Average Linkage Agglomerative Clustering

Average Linkage Agglomerative Clustering Solved Example

- Let us compute the distance of **{a, c, e}** from other clusters.
- $d(\{a, c, e\}, \{b, d\}) = \text{avg}\{d(a, b), d(a, d), d(c, b), d(c, d), d(e, b), d(e, d)\}$
- $d(\{a, c, e\}, \{b, d\}) = \frac{9+6+7+9+10+8}{3*2} = 8.16$

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

	a,c,e	b,d
a,c,e	0	8.16
b,d	8.16	0

Average Linkage Agglomerative Clustering

Average Linkage Agglomerative Clustering Solved Example

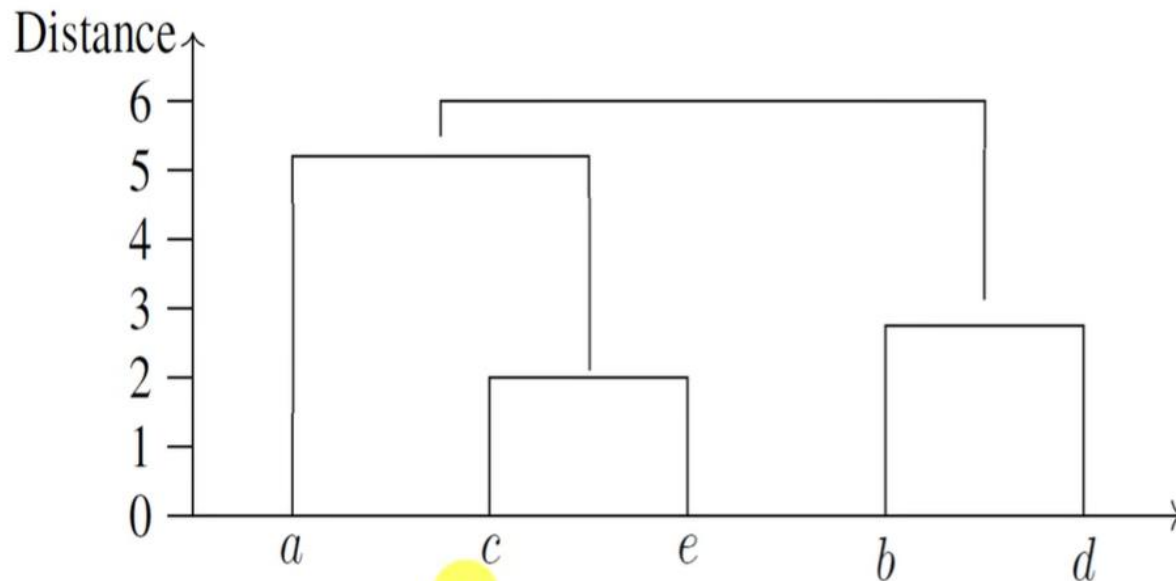
- Let us compute the distance of {a, c, e} from other clusters.

	a,c,e	b,d
a,c,e	0	8.16
b,d	8.16	0

- $d(\{a, c, e\}, \{b, d\}) =$
 $avg\{d(a, b), d(a, d), d(c, b), d(c, d), d(e, b), d(e, d)\}$
- $d(\{a, c, e\}, \{b, d\}) = \frac{9+6+7+9+10+8}{3*2} = 8.16$
- Only two clusters are left.
- We merge them form a single cluster containing all data points.

Average Linkage Agglomerative Clustering

Average Linkage Agglomerative Clustering Solved Example



	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

THANK YOU